

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Mechatronics Engineering **Lab Report**

Course Title : Power Electronics and Drives
Course Code : Sessional: MTE 3201
Experiment No. : 04
Experiment Name : Study of Thyristor (SCR) Characteristics.

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Experiment No: 04

Experiment Name: Study of Thyristor (SCR) Characteristics.

Objectives:

1. To examine how the gate signal controls the turn-on of a thyristor (SCR).
2. To find the gate trigger voltage (V_{GT}) at which the device starts conducting.
3. To compare the anode-to-cathode voltage (V_{AK}) of the device before and after it is triggered.

Introduction:

The thyristor, also called a silicon controlled rectifier (SCR), is a four-layer (PNPN), three-terminal semiconductor switch. With the anode made positive relative to the cathode, the device still stays in its forward-blocking (off) state and passes only a negligible leakage current until a suitable pulse is injected into the gate. Raising the gate-to-cathode voltage drives a growing gate current, and when it reaches the gate trigger level (V_{GT}/I_{GT}) the device snaps into the forward-conduction (on) state. At that instant V_{AK} drops from almost the whole supply voltage to a small on-state drop of about one volt. The SCR then keeps conducting even after the gate drive is taken away and turns off again only when the anode current falls under the holding current. In this experiment the gate voltage was raised in steps with a potentiometer while a storage oscilloscope displayed V_{AK} , so that the gate voltage at which triggering takes place could be located.

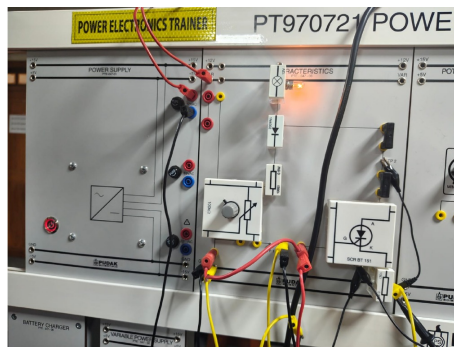


Figure-01: PT970721 Power Electronic Trainer

Required Apparatus:

1. Power supply unit, PTE-047-01
2. Thyristor characteristics module, PTE-047-03
3. AC lamp, 12 V / 2 W
4. Diode, 1N4002
5. Potentiometer, 10 k Ω
6. Thyristor (SCR), BT151
7. Digital multimeter
8. Storage oscilloscope
9. Connecting leads

Circuit Diagram:

The circuit that was wired on the trainer is shown in Figure-02. It is reproduced from the hand-drawn sketch made in the laboratory during the experiment.

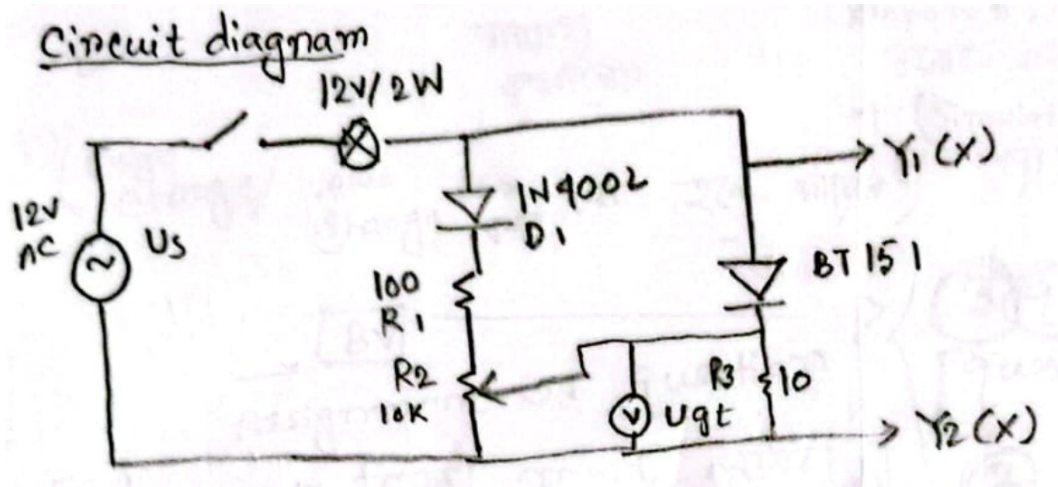


Figure-02: SCR gate-triggering circuit ($R_1 = 100 \Omega$, $R_2 = 10 \text{ k}\Omega$, $R_3 = 10 \Omega$)

Principle:

A 12 V AC source feeds the circuit through a 12 V / 2 W lamp, which serves as a visible load and limits the current. After the lamp the circuit divides into two paths. One path goes through the BT151 thyristor and the small sensing resistor R_3 . The other goes through the 1N4002 diode D_1 , the fixed resistor R_1 and the potentiometer R_2 ; this path produces a rectified, adjustable voltage. The slider of R_2 is joined to the thyristor gate, so turning the knob changes the gate drive. The voltage across R_3 is the gate voltage U_{GT} , and the voltage across the thyristor is U_{AK} . Both are displayed together on the oscilloscope, with CH1 on U_{AK} and CH2 on U_{GT} .

When the slider is near the low end, U_{GT} is tiny and the gate current is too weak to start regeneration inside the PNP structure. The SCR therefore keeps blocking, and the anode-cathode voltage follows the supply waveform (the device is *not triggered*). As U_{GT} is raised past the minimum gate trigger voltage, the device latches on, U_{AK} falls to the small conduction drop, and the load starts drawing current (the device is *triggered*). Noting U_{GT} and U_{AK} at several potentiometer positions therefore gives an approximate value of the gate trigger voltage.

Procedure:

1. The power supply, thyristor-characteristics module and load panel were mounted on the trainer and interconnected following the circuit diagram of the lab manual.
2. The AC source, lamp, diode D_1 , resistor R_1 , potentiometer R_2 , resistor R_3 and the BT151 were joined exactly as drawn in Figure-02.
3. Oscilloscope channel CH1 was placed across the anode and cathode to display U_{AK} , and channel CH2 was placed across R_3 to display U_{GT} .
4. Potentiometer R_2 was turned to its minimum position and the supply was then switched on.
5. The knob was turned up slowly, and at each position U_{GT} and the matching U_{AK} were read from the oscilloscope.
6. The position at which the SCR moved from the blocking state to the conducting state was identified from the abrupt change in the U_{AK} trace.
7. All readings were entered in the data table, and screenshots of the oscilloscope were taken at the different stages.

Data Table:

No.	U_{GT} (Gate Voltage)	U_{AK} (Anode-Cathode Voltage)	$I_G = U_{GT}/R_3$	Remarks
1	26 mV	3.64 V	2.6 mA	Not triggered
2	182 mV	2.90 V	18.2 mA	Triggered
3	224 mV	3.68 V	22.4 mA	Triggered
4	186 mV	2.06 V	18.6 mA	Triggered
5	184 mV	3.20 V	18.4 mA	Triggered

Note: CH1 measured U_{AK} and CH2 measured U_{GT} , as shown in the circuit diagram. Only reading No. 1 left the thyristor untriggered; every later, higher gate-voltage setting turned it on. The gate current column is an added estimate, obtained by dividing U_{GT} by the $10\ \Omega$ sensing resistor R_3 .

Result:

Figure-03 collects the oscilloscope screenshots taken at the various stages of the experiment. yellow trace is U_{AK} and the cyan trace is U_{GT} The .

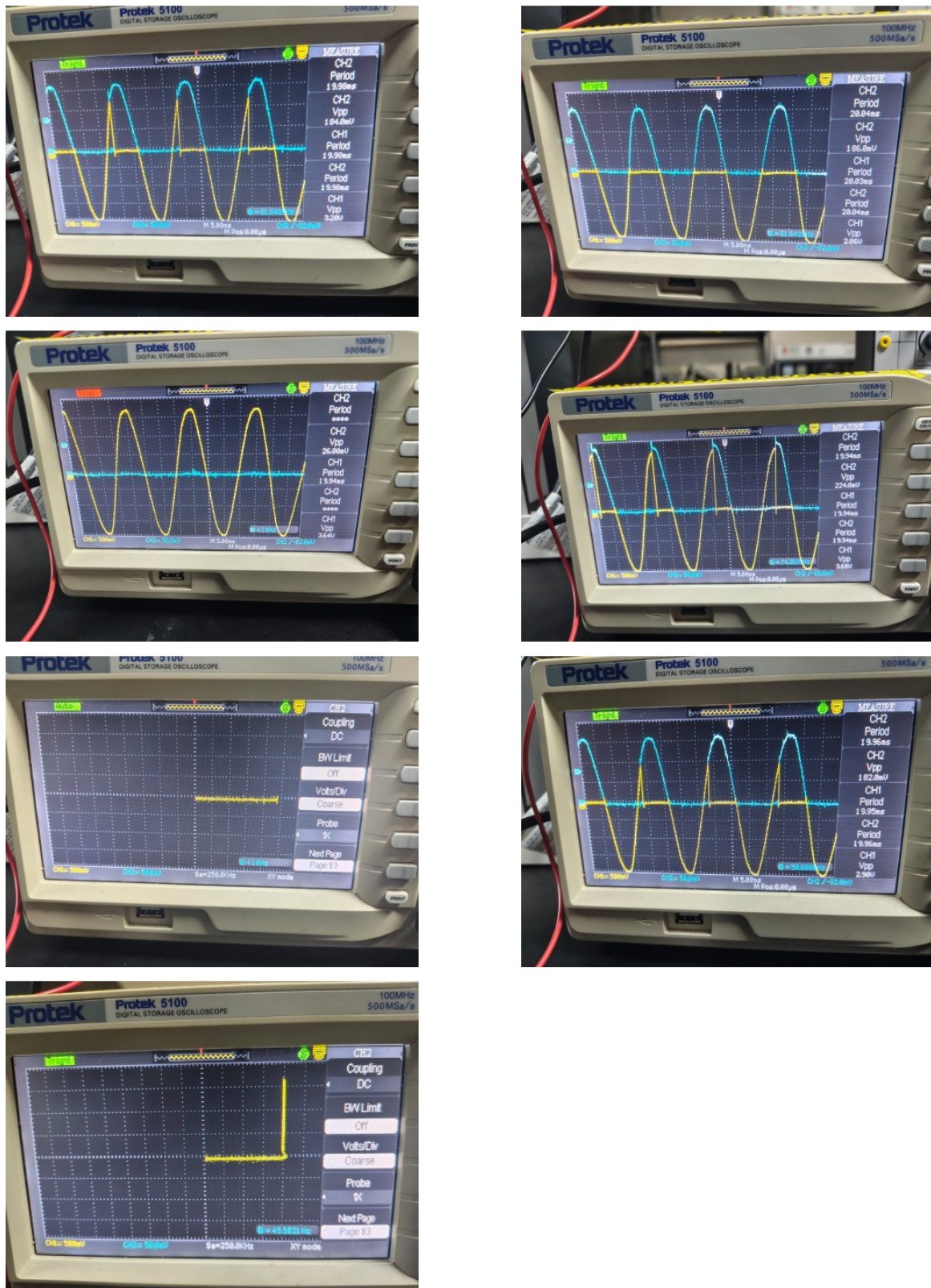


Figure-03: Oscilloscope captures at different stages of the experiment

Discussion:

The gate-triggering behaviour of the BT151 SCR was examined by increasing the gate voltage U_{GT} with a potentiometer and watching the anode-cathode voltage U_{AK} on a storage oscilloscope. With the lowest setting ($U_{GT} = 26$ mV, corresponding to roughly 2.6 mA through R_3) the gate drive was inadequate, so the device stayed in the forward-blocking state and was recorded as not triggered. Once the potentiometer was moved on, the gate voltage lay between about 182 mV and 224 mV (roughly 18 to 22 mA), and the SCR turned on at every one of these settings.

The measurements therefore place the minimum trigger voltage between 26 mV and 182 mV; the lowest value that actually fired the device, 182 mV, is the best upper estimate obtained. A finer sweep between these two settings would be needed to narrow it down. The U_{AK} figures of the triggered readings (2.06 V to 3.68 V) are scattered, and the 3.64 V of the untriggered reading is not clearly different from them. This is understandable because a single meter-style number depends on the point of the AC cycle at which the SCR fires, on the drop across the lamp and the diode branch, and on reading tolerances. For this reason the change of state was decided from the shape of the U_{AK} waveform on the oscilloscope rather than from the numbers alone.

Conclusion:

The gate-triggering characteristic of the thyristor was studied successfully. The SCR blocks in the forward direction as long as the gate voltage stays under a minimum trigger value; when this value is exceeded it changes rapidly to the conducting state and its anode-cathode voltage drops sharply. The experiment confirmed the basic switching action of the SCR and gave an approximate gate trigger voltage of about 182 mV (about 18 mA of gate current) for this circuit.

References:

- [1] Laboratory manual, *Power Electronics and Drives Sessional (MTE 3201)*, Dept. of Mechatronics Engineering, RUET.
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- [4] P. S. Bimbhra, *Power Electronics*, 5th ed. New Delhi, India: Khanna Publishers, 2012.
- [5] NXP/WeEn Semiconductors, “BT151 series, Thyristors (SCR), 12 A, 500-800 V,” product datasheet.